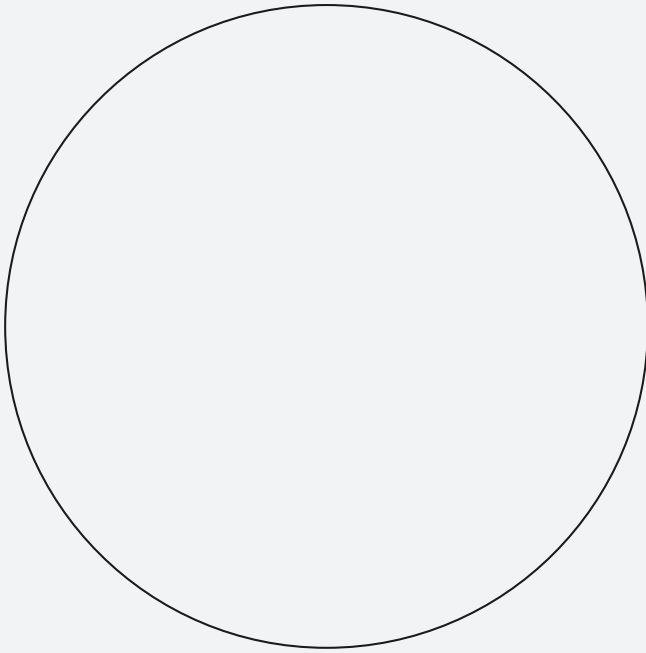


Leviathan Validator,
“3.0 Mainnet Version”
Audit



Code by Optim Labs
Audit by ESYS Labs

The audit takes into account the files **account.ak**, **always.ak**, **intent.ak**, **types.ak**, **utils.ak**, and **validation.ak**, with the final resulting and approved to be deployed code provided at commit SHA **6cc2516f83c265a52dd76d074b74da15993f710c**, provided in a self-contained repository that contains a nix shell environment to compile reproducibly.

We have worked with Optim Labs in a fix-by-found manner until all critical, major and minor issues were removed, and, at the request of Optim Labs we have produced this report on the final provided to public version, which has undergone a secondary audit to ensure that no issues remained. We do not assess optimization, only correctness of the code.

Informational issues remain listed.

1. Informational

The Intent Shown in Wallet is not Clear Text

Line 62 of **intent.ak**.

Messages are shown padded in bytes. Fix on Wallet side available.

2. Informational

Sequencer can eject all accounts without signatures

Line 69 of **validation.ak**.

During a sync call the system is blind on whether the output address is the owner pubkey or owner Leviathan account, thus allowing the sequencer to eject anyone and everyone at will from their account.

3. Informational

Exchange intent can be matched arbitrarily many times partially

Line 598 of **validation.ak**.

There is no option to set a minimum match size per market, and thus the only minimum is one token.

4. Informational

Very unlikely hash collision

Line 74 of utils.ak .

One byte is being dropped during hash equality checking. This can cause a hash collision.

5. Informational

Utxo Age is not enforced onchain

Line 76 of validation.ak .

The age of the utxo to be spent initially is left only to the sequencer. During initial bridging it should be enforced that the utxo age is final.
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